MIL-I-27203B(USAF)
29 December 1987
SUPERSEDING
MIL-I-27203A(USAF)
25 May 1965

MILITARY SPECIFICATION

INDICATOR, POSITION, WING FLAP

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for 28V dc integrally lighted wing flap position indicators.

1.2 Classification. Indicators shall consist of the following types, as specified (see 6.2):

ALU-3/A	Cover glass with antireflection coating per MIL-C-675
ALU-3A/A	Cover glass with multilayer antireflection coating of high efficiency (superior to coating specified for ALU-3/A)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

Military

MIL-P-116	Preservation, Method Of
MIL-C-675	Coating Of Glass Optical Elements (Anti-Reflection)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6610
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Military

MIL-T-5583	Transmitter, Position, 28-Volt DC
MIL-I-7062	Indicators, Position, Control Surfaces, 28 volt DC, General Specification For
MIL-T-26609	Transmitter, Position
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification For

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-794	Parts And Equipment, Procedures For Packaging And Packing Of
MIL-STD-810	Environmental Test Methods Engineering Guideslines
MS24515	Lamp, Incandescent, T-1, Sub-miget Flanged
MS24550	Transmitter, Position
MS25237	Lamp, Incandescent, Single Contact Midget Flange Base (T-1-3/4 Bulb)
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft

Air Force-Navy Aeronautical

AN5786	Transmitter & Position, 28 Volt DC
AN5787	Transmitter & Position, Gearless

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise speciifed, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society For Testing And Material (ASTM)

ASTM D3951 Packaging, Commercial

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(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organization which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict betwen the text of this specification and the reference cited herein (except for associated detail specifications, specification sheets or Ms standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Indicator furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.5 and 6.5).

3.2 General specification. The requirements of MIL-I-7062 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.3 Design and construction. The indicator shall be designed to operate with position transmitter conforming to MIL-T-5583 and MIL-T-26609. The pointer shall travel approximately 180 mechanical degrees for 90° rotation of an AN5786 or MS24550 transmitter or for 180° rotation of an AN5787 transmitter.

3.3.1 Resistance. The direct-current resistance of the indicator measured between connector pins, A, B, and C shall be 475 $\pm$100 ohms. Deviation from the average of the resistance values between the 3 coils shall not exceed $\pm$1.50 percent.

3.3.2 Reliability. The indicator shall have a minimum mean-time-between-failures (MTBF) of 225 hours at a 90 percent confidence level.

3.3.3 Vibration. The indicator shall be designed to operate for not less than 21,000 cycles while being subjected to vibration up to 2g.

3.4 Case. The case shall be in accordance with MS33639, for the 2-inch size, except as shown on figure 1. The case shall provide a hermetically sealed inclosure for the internal mechanism of the indicator.

3.4.1 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be either 100 percent helium or a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere.

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3.4.2 Cover glass position. The distance between the outer surface of the cover glass and the front edge of the case shall be the minimum practicable and shall not exceed .031 inch.

3.4.3 Fogging. There shall be no evidence of moisture or oil fogging of the cover glass when the indicator is subjected to a temperature of 85° ± 5°C for a minimum period of 2 hours and then the face of the cover glass immediately immersed in an ice bath for 3 minutes.

3.5 Dial. The dial shall conform to figure 1. It shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen nor turn when the indicator is vibrated. The diameter of the dial shall be not less than .750 inches when measured across the outside ends of the graduations.

3.5.1 Markings. The dial markings shall be as follows and shall be durable to withstand usage encountered in service. The colors shall be in accordance with FED-STD-595.

Markings	Height or Length Inches ± .016	Width of Line or Graduation Inches ±0.005	Finish
Numerals 20, 40, 60, 80 and letterings UP	0.156	0.020	37875
Lettering FLAPS	0.125	0.020	37875
Lettering DOWN	0.109	0.020	37875
LETTERING PERCENT EXTENDED	0.094	0.020	37875
Major graduations	0.188	0.031	37875
Minor graduations	0.094	0.020	37875
Shaded portion of pointer	----	-------	37875
Background of dial	----	-------	37038
Lettering ALU-3/A or ALU-3A/A	0.047	-------	37038

3.5.2 The dial markings shall include an angle of 180° ±15' and shall conform to the following:

Dial indication	UP	10	20	30	40	50	60	70	80	90	DN
Degress from vertical θ	0	18	36	54	72	90	108	126	144	162	180

3.6 Reflective surfaces (ALU-3A/A). All reflecting glass surfaces shall be provided with a reflection-reducing coating which meets the requirements of MIL-C-675, in addition to withstanding the environmental conditions specified herein, except the following reflectance tolerances shall apply:

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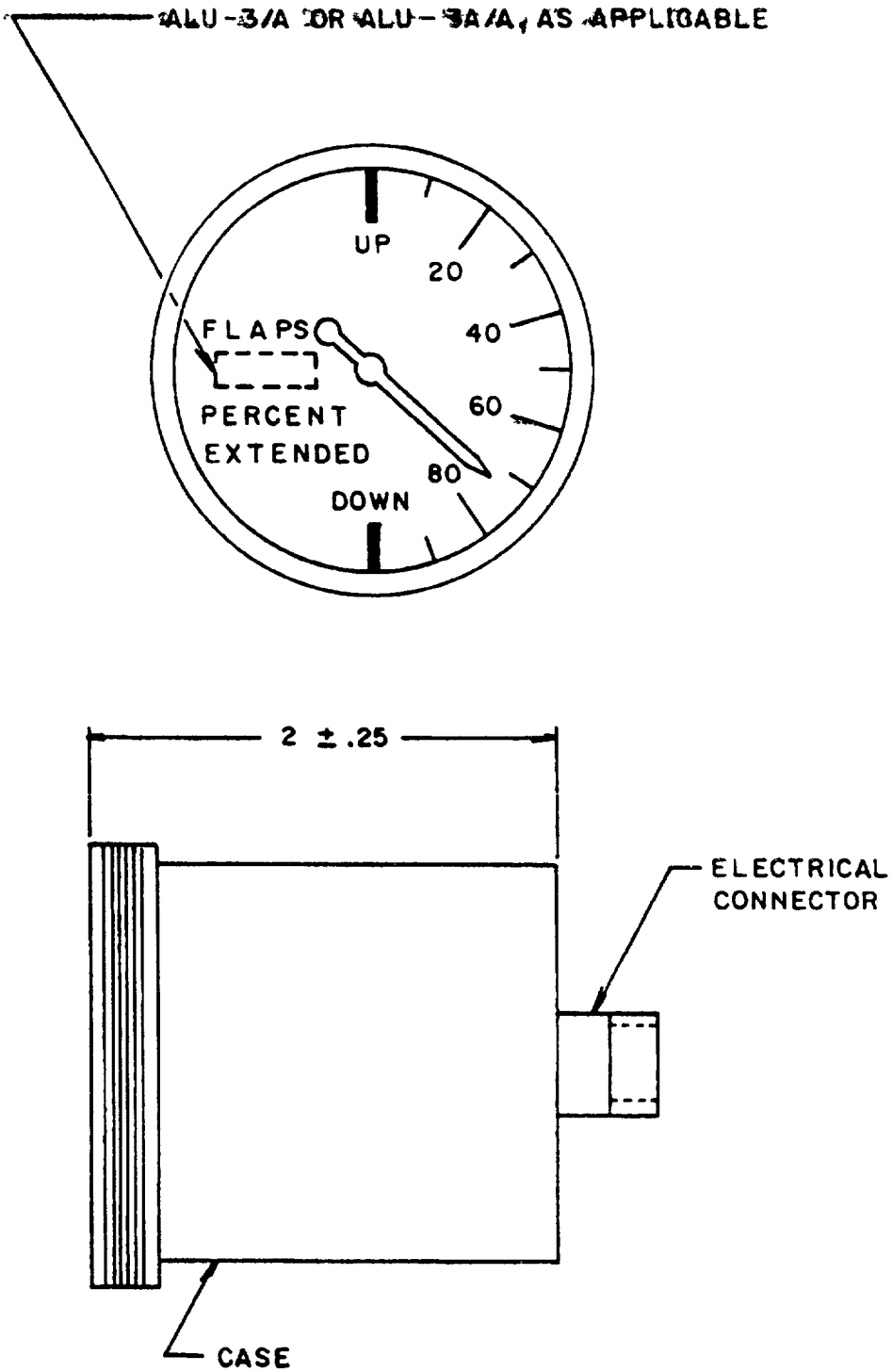


FIGURE 1. Dial and case.

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Angle of Incidence	Wavelength (Millimicrons)	Percent Reflectance
0°	450-675	0.6 maximum
0°	425-700	0.5 average
30°	450-625	1.0 maximum
30°	425-700	0.5 average

3.7 Pointer overlap. The pointer length shall be such that the pointer tip will overlap the shortest graduation by 0.20 of its length $\pm$1.062 inch.

3.8 Electrical connector. Unless otherwise specified, the electrical connector shall be a Bendix Type PTIH-10-6P, or equal. The connector shall be provided with a cap or cover and gasket to prevent penetration of water vapors.

3.9 Integral lighting. The indicator lighting shall be in accordance with MIL-L-27160. The lighting system shall be so designed that replacement lamps may be accurately positioned.

3.9.1 Lamps. The light source for the indicator shall be three or more MS25237 lamps or lamps in accordance with MS24515 having a metal mounting base.

3.9.1.1 Lamp positions. The lamp shall be located within the hermetically sealed inclosure.

3.9.1.2 Lamp replacement. Lamp replacement shall be accomplished at overhaul depots.

3.10 Identification of product. Equipment, assemblies, and parts shall be marked in accordance with MIL-STD-130.

3.11 Recycled and reclaimed materials. The use of recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purhcase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to asure supplies and services conform to prescribed requirements.

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4.1.1 *Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specifications shall become a part of the contractor's overall inspection system or quality program. The absence of any inspction requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 General specification. The inspection and testing of indicators shall be in accordance with MIL-I-7062 except as specified herein.

4.3 Test conditions

4.3.1 Identification of tests. For the purpose of this specification, identification of tests shall be as follows:

MIL-I-27203	MIL-I-7062
Acceptance tests	Inspection tests
Sampling plan A tests	Sampling tests
Sampling plan B tests	Special tests
Qualification tests	Qualification tests

4.3.2 Scale error. The scale error shall be read from the test transmitter scale with the indicator pointer on each major graduation of the indicator dial. The scale error shall not exceed 4 mechanical degrees.

4.3.2.1 For the scale error test, two indicators and one transmitter shall be used. The scale shall be within 1.25 times the scale error specified in 4.3.2. The wiring shall be as shown on figure 2.

4.4 Test methods

4.4.1 Acceptance tests.

4.4.1.1 Individual tests. The indicator shall be subjected to the individual tests specified in MIL-L-27160 in addition to MIL-I-7062 requirements.

4.4.2 Sampling plan A tests. The indicator shall be subjected to the sampling plan A tests specified in MIL-L-27160 in addition to MIL-I-7062 requirements.

4.4.2.1 Vibration error. The indicator shall be attached to a rigid fixture capable of transmitting the vibration conditions specified herein and mounted on the vibration stand in the normal operation position.

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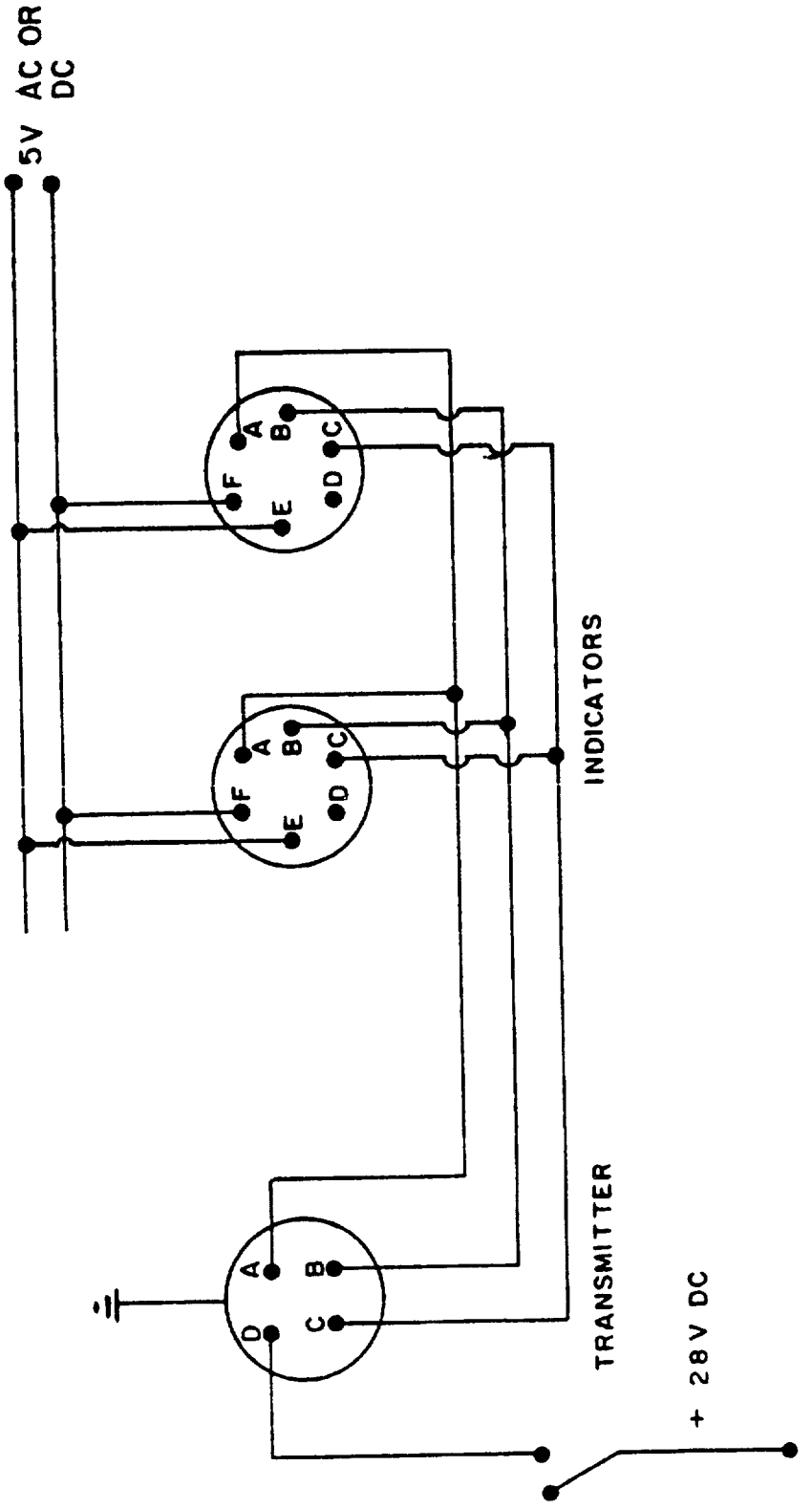


FIGURE 2. External wiring diagram.

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With the lighting system energized with 5V dc, the indicator shall be energized to 28V dc and subjected to a vibration test in accordance with MIL-STD-810, equipment category b-1, part 2 of procedure 1, curve B, except that from 5 to 55 Hz, the displacement shall be 0.010 inch and from 300 to 500 Hz, the g-level shall be 1.0g in lieu of 2.0g. From 55 to 300 Hz, the g-level shall be 2g. Before vibration is started, the position transmitter shall be adjusted so that the indicator pointer will be at the 0° position and shall not be readjusted after vibration is started. During vibration, the maximum pointer oscillation shall not exceed 10 mechanical degrees double amplitude and the pointer shaft (measured from the center of pointer oscillation) shall not exceed 4 mechanical degrees from the original 0° position settings. All lamps shall be operating at the completion of this test.

4.4.3 Sampling plan B tests. The indicator shall be subjected to the sampling plan B tests specified in MIL-L-27160 in addition to MIL-I-7062 requirements.

4.4.3.1 Vibration failure and cycling. The three sample indicators shall be subjected to a combination vibration and pointer cycling test for a period of 115 hours. A vibration machine shall be used that will provide a double amplitude of 0.005 to 0.02 inch at a frequency of 45 to 60 Hz. The amplitude and frequency chosen shall provide a vibration level of 1.5g $\pm$ percent. The direction of vibration shall be perpendicular to the longitudinal axis of the indicator. Before starting this test, the indicators shall be subjected to the scale and friction error tests. Pointer cycling shall consist of pointer motion from the up position (0 percent extended) to the down position (100 percent extended) and back to the up position. This deflection shall constitute 1 cycle and shall be repeated for a period of 115 hours at a rate of not less than 3 nor more than 4 cpm. The vibration specified shall be applied for 15 minutes during each hour. Upon completion of this test, the indicators shall again be subjected to the scale and friction error tests. The changes in scale and friction error at the various test positions shall not exceed 2 mechanical desgrees. There shall be no looseness in the mechanism nor damage to any part of the indicators as a result of this test.

4.4.3.2 Reliability. Satisfactory completion of all the tests specified herein, with no failures, will demonstrate an MTBF of 225 hours at a 90 percent confidence level.

4.4.3.3 Fogging. The indicator shall be placed in oven at a controlled ambient temperature of $85^{o} \pm 5^{o}C$ for minimum period of 2 hours. Within 10 seconds after removing the indicator from the oven, the indicator face shall be immersed to a depth of between .250 and .333 inch in an ice bath maintained at a temperature between 0° and $5^{o}C$. After 3 minutes, the indicator shall be removed from the ice bath and quickly wiped dry (compressed air shall not be used). The cover glass and dial shall be inspected immediately for fogging or damage. (Any evidence of fogging or damage shall be cause for rejection.)

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4.4.4 Reflectance (ALU-3A/A). The reflectance of all reflecting glass surfaces shall be measured and the reflectance tolerances shall not exceed those specified in 3.6.

4.5 Qualification testing

4.5.1 Test samples. The test samples shall consist of three indicators representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as requied by the contracting activity.

4.5.2 Qualification tests. The qualification tests shall consist of all the tests specified herein.

4.6. Inspection of packaging. The inspection of the packaging and interior package marking shall be in accordance with group A and B quality conformance inspection requirements, section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements MIL-STD-129.

5. Packaging

5.1 Preservation. Preservation shall be level A, C, or Industrial as specified. (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Indicator shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quanitity unit packs or one each in accordance with Method IC-1 of MIL-P-116. Each indicator shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of indicator shall be in accordance with ASTM D3951.

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5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistant with the protection required.

5.2.2 Level B. Indicator packaged as specified in 5.1.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicator shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2) interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The indicator covered by this specification is intended for use in indicating the position of the wing flaps in percent of travel from the up position and for use where a white integrally lighted indicator is required. The position transmitters used with this indicator are connected to the flap actuating mechanism so that the transmitter shaft is rotated 90 or 180 mechanical degrees, as applicable, for 100-percent flap travel. The indicator pointer rotates approximately 180° to indicate the full-down position.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Type of indicator required (see 1.2)

c. When sampling plan B tests will not be conducted.

d. Levels of packaging and packing required and reinspection date marking

6.3 Clamp mounting. The indicator will be installed by clamp mounting.

6.4 Definition

6.4.1 Double amplitude. Double amplitude is the extent of motion as measured from one extreme to the opposite extreme.

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6.5 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The acitvity responsible for the Qualified Products List is the Oklahoma City Air Logistics Center/MMEDO, Tinker AFB, OK 73145-5990 and information pertaining to qualification of products may be obtained from that activity.

6.6 Subject term (key word) listing.

ALU-3
Indicator
Integrally lighted
Position
Swing flap
28 V dc

6.7 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with repect to the previous issue due to the extensiveness of the changes.

Custodian:
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Preparing Activity:
AIR FORCE -71

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b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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